

Particles and Nuclei 2

- 2/3/26
- Announcements:
- HW 2 Due a Wk from Thurs 2/12/26
- Some reading for skimming thru Griff Ch 6 beginning

Outline

- Finish exercise problem from class last time
- Commentaries on some important invariants
- More frame/beam scenarios

Exercise

- Two particles a with mass m collide. One is at rest in the lab. The other has beam energy E and collides with it and produces two heavier particles, each of mass M .
- $a+a \rightarrow b+b$;
- Half the class find expression for four vector p_1 dot p_2 and p_1+p_2 in lab frame “(fixed target)”.
- Half the class find “ ” in the COM frame

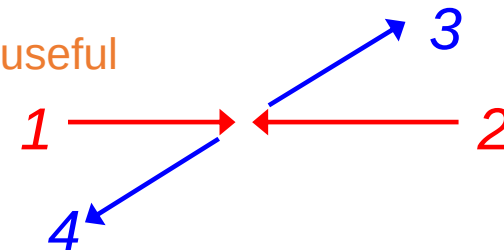
In class we mostly focused on the initial state (p_1+a)
For Lab frame it was easiest to evaluate p_1 dot p_2 directly
For the COM frame p_1+p_2 (four vector then square it)

The next slides show this done in a different context, at the end of class, I set the two results equal to demonstrate the advantage of using colliders (ie COM frame) or conversely, the much larger beam E needed for the same COM energy in the fixed target

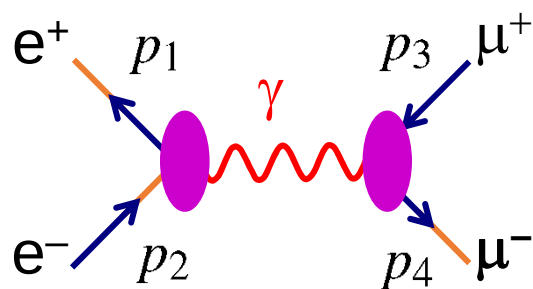
Mandelstam s , t and u

- ★ In particle scattering/annihilation there are three particularly useful Lorentz Invariant quantities: s , t and u

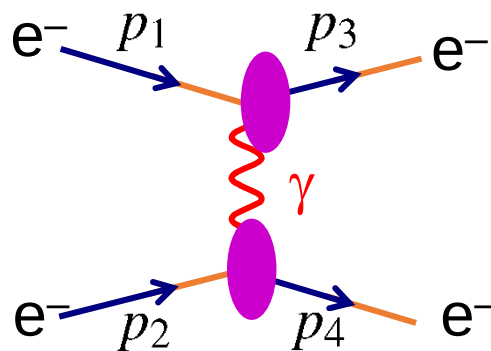
- ★ Consider the scattering process $1 + 2 \rightarrow 3 + 4$



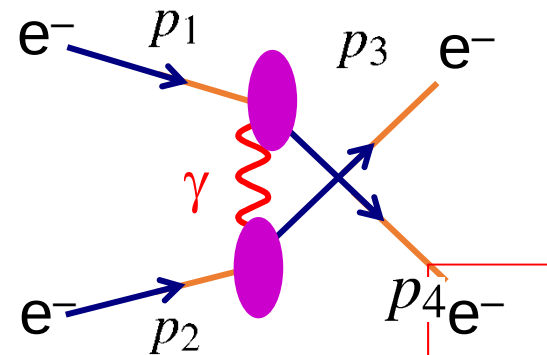
- ★ (Simple) Feynman diagrams can be categorised according to the four-momentum of the exchanged particle



s-channel



t-channel



u-channel

- Can define **three** kinematic variables: s , t and u from the following four vector scalar products (squared four-momentum of exchanged particle)

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2$$

u, t, s – cross section predictions

- Perturbative cross sections ... also NLO often given

$$\frac{d\sigma}{d\Omega}(qq \rightarrow qq) = \frac{\alpha_s^2}{9s} \left[\frac{t^2 + s^2}{u^2} + \frac{s^2 + u^2}{t^2} - \frac{2s^2}{3ut} \right]. \quad (50.7)$$

Annihilation of e^+e^- into $\gamma\gamma$ has the cross section

$$\frac{d\sigma}{d\Omega}(e^+e^- \rightarrow \gamma\gamma) = \frac{\alpha^2}{2s} \frac{u^2 + t^2}{tu}. \quad (50.8)$$

The related QCD process also has a triple-gluon coupling. The cross section is

$$\frac{d\sigma}{d\Omega}(q\bar{q} \rightarrow gg) = \frac{8\alpha_s^2}{27s} (t^2 + u^2) \left(\frac{1}{tu} - \frac{9}{4s^2} \right). \quad (50.9)$$

The crossed reactions are

$$\frac{d\sigma}{d\Omega}(qg \rightarrow qg) = \frac{\alpha_s^2}{9s} (s^2 + u^2) \left(-\frac{1}{su} + \frac{9}{4t^2} \right) \quad (50.10)$$

and

$$\frac{d\sigma}{d\Omega}(gg \rightarrow q\bar{q}) = \frac{\alpha_s^2}{24s} (t^2 + u^2) \left(\frac{1}{tu} - \frac{9}{4s^2} \right). \quad (50.11)$$

Finally,

- 1987 Paper Photon production

$$gg \rightarrow \gamma g \quad -\frac{e_s^2}{3} \left[\frac{u}{s} + \frac{s}{u} \right]$$

$$q\bar{q} \rightarrow \gamma g \quad \frac{8}{9} e_q^2 \left[\frac{u}{t} + \frac{t}{u} \right]$$

$$q\bar{q} \rightarrow \gamma\gamma \quad \frac{2}{3} e_q^4 \left[\frac{t}{u} + \frac{u}{t} \right]$$

$$\begin{aligned}
 gg \rightarrow \gamma\gamma \quad & \left[\sum_{i=1}^{n_f} e_q^2 \right]^2 \frac{\alpha_s^2}{8\pi^2} \left\{ \frac{1}{8} \left[\left[\frac{s^2+t^2}{u^2} \ln^2 \left[-\frac{s}{t} \right] + 2 \frac{s-t}{u} \ln \left[-\frac{s}{t} \right] \right]^2 \right. \right. \\
 & + \left[\frac{s^2+u^2}{t^2} \ln^2 \left[-\frac{s}{u} \right] + 2 \frac{s-u}{t} \ln \left[-\frac{s}{u} \right] \right]^2 + \left[\frac{t^2+u^2}{s^2} \left(\ln^2 \frac{t}{u} + \pi^2 \right) + 2 \frac{t-u}{s} \ln \frac{t}{u} \right]^2 \Big\} \\
 & \times \frac{1}{2} \left[\frac{s^2+t^2}{u^2} \ln^2 \left[-\frac{s}{t} \right] + 2 \frac{s-t}{u} \ln \left[-\frac{s}{t} \right] + \frac{s^2+u^2}{t^2} \ln^2 \left[-\frac{s}{u} \right] + 2 \frac{s-u}{t} \ln \left[-\frac{s}{u} \right] \right. \\
 & + \left. \frac{t^2+u^2}{s^2} \left[\ln^2 \frac{t}{u} + \pi^2 \right] + 2 \frac{t-u}{s} \ln \frac{t}{u} \right] \\
 & + \frac{\pi^2}{2} \left[\left[\frac{s^2+t^2}{u^2} \ln \left[-\frac{s}{t} \right] + \frac{s-t}{u} \right]^2 + \left[\frac{s^2+u^2}{t^2} \ln \left[-\frac{s}{u} \right] + \frac{s-u}{t} \right]^2 \right] + 4 \Big\}
 \end{aligned}$$

Mandelstam s , t and u

- From current pset

$$s = 4(p^2 + m^2 c^2)/c^2$$

$$t = -2p^2(1 - \cos \theta)/c^2$$

$$u = -2p^2(1 + \cos \theta)/c^2$$

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Problem 3.26

$$t = (p_1 + p_3)^2$$

$$\sim 2p_1 \cdot p_3 = E_1 E_3 - \vec{p}_1 \cdot \vec{p}_3$$

$\swarrow \propto \cos\theta$

Most important: s (?)

- Generally the COM mass ENERGY of the colliding beams is what is quoted – this is $\sqrt{s}$ [cause s, t, u are 4- v 's squared]
- B.c. “Universal” average behavior and possible kinematic final states is determined by the INITIAL STATE Total E/p
 - In other words observed physics should be same only for same COM E ($\sqrt{s}$)
 - Even for t/u channel scattering. generally, esp. for hadronic scattering
 - can depend on what physics (t can be too)
- Same $\sqrt{s}$ $\rightarrow$ “Reproducibility”

INSPIRE HEP literature find cn cms

Published in: JINST 19 (2024) 01, C01049

pdf DOI cite claim reference search 0 citations

Nonresonant central exclusive production of charged-hadron pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

TOTEM and CMS Collaborations • Aram Hayrapetyan (Yerevan Phys. Inst.) et al. (Jan 25, 2024)

e-Print: 2401.14494 [hep-ex]

Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the emu final state at $\sqrt{s} = 13$ TeV within CMS

CMS Collaboration • Daina Leyva Pernia for the collaboration. (Jan 22, 2024)

Published in: PoS EPS-HEP2023 (2024) 489 • Contribution to: EPS-HEP2023, 489

pdf DOI cite claim reference search 0 citations

High Energy Physics - Experiment

arXiv:2204.12899 (hep-ex)

[Submitted on 27 Apr 2022 (v1), last revised 7 Mar 2023 (this version, v2)]

Improving constraints on gluon spin-momentum correlations in transversely polarized protons via midrapidity open-heavy-flavor electrons in $p^\uparrow + p$ collisions at $\sqrt{s} = 200$ GeV

find cn phenix

Identified charged-hadron production in $p+Al$, ^3He+Au , and $Cu+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV and in $U+U$ collisions at $\sqrt{s_{NN}} = 193$ GeV

PHENIX Collaboration • N.J. Abdulameer (Debreceen U.) et al. (Dec 14, 2023)

e-Print: 2312.09827 [nucl-ex]

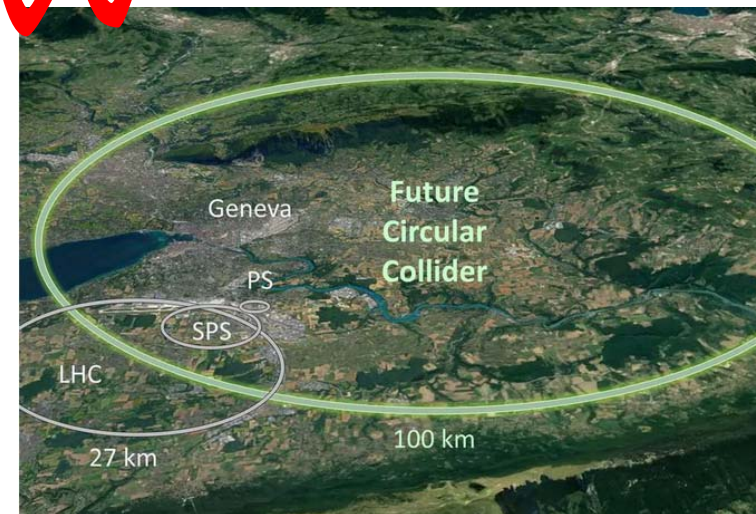
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HEP → Future / 10 TeV “pCM”

- Higg's factories Higgs discovered 2012 Mass 125 GeV
- Nominally Ebeam $125/2 = 62.5$ GeV → RHIC ?
 - But for protons like at LHC/RHIC quarks inside only carry small fraction of Ebeam.
- “pCM” : ‘parton’ CM energy → Future Circular Collider
~100 TeV proton CM
- Use Lepton: previous p5 high priority :
 - Long time: ILC – International Linear Collider $e^+/-$ no Bremstrahlung
 - LEP: (former LHC): 100 GeV e beams → circular doable – China!
- MUON COLLIDER!
 - Like radioactive nucl. beam/ FRIB
- Plasma Wake acceleration
- [COLLIDERS THOUGH!]
 - Relativity : “fixed target → $\sqrt{E_{\text{beam}}}$ ”

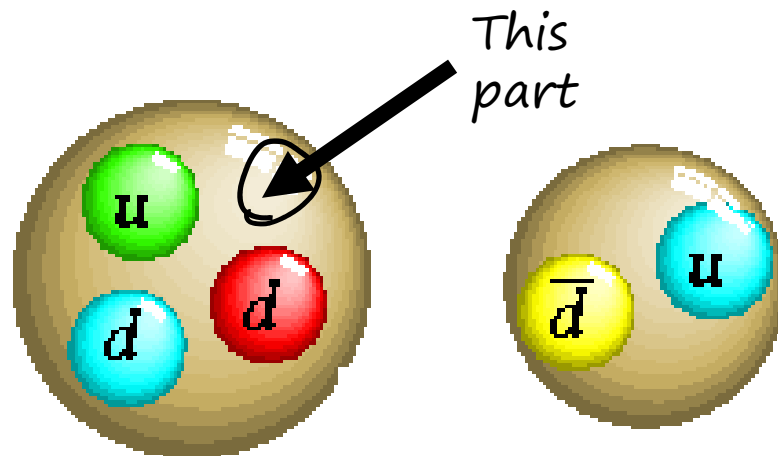
Review

Recommendation 4: Invest in a comprehensive initiative to develop the resources—theoretical, computational, and technological—essential to realizing our 20-year strategic vision. This includes an aggressive R&D program that, while technologically challenging, could yield revolutionary accelerator designs that chart a realistic path to a 10 TeV pCM collider.

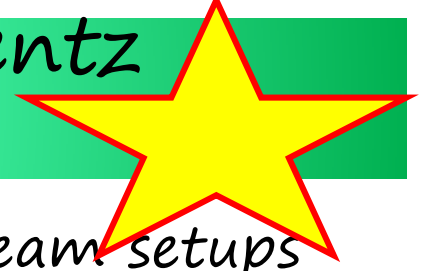


pcm - s ... Nonperturb QCD

- PS in hadron+hadron we call that pCM also $\hat{s}$
- Another reason for s importance is because of the “separation of scales” (energy scales)
 - Parton-Parton COM may be relevant for production of Higg's, new particles
 - But for the non-perturbative QCD hadronic physics (the parton's packaging) we shall see the description of that physics can be FRAME DEPENDENT
 - So most logically we use the COM $E(s)$ of the HADRONS to try our best to UNIVERSALIZE THAT
- Also why even for t scattering



Experimental Reality : Lorentz Transformations



- Particle physics actually done with variety of beam setups
 - 1 beam, fixed target
 - 2 beams colliding same energy
 - 2 beams colliding different energies
- Best to describe physics in ways that don't depend on which setup is used
- But we must often compare results that should be ~same physics but obtained with different beam setups.
- THUS Lorentz Transformations Necessary
- (e.g. for *reproducibility*)
- Many textbooks on QM describe only FIXED target case
- Theorists generalize results with Lorentz invariants, often " s , t , u " (Mendallstaam vars)

Exercise (last time)

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CoM (yes $\equiv$ Collider) frame s

- Both result in super trivial stuff – use E_b for beam E
 - Also as long as E 's are big enough doesn't really need to be same mass particles

- 1) Add vectors first :

$$s = \left([\vec{E}_b, \vec{p}_1] + [\vec{E}_b, \vec{p}_2] \right)^2 = \left([2E_b, 0] \right)^2 = (2E_b)^2 = 4E_b^2$$

- 2) Do 4-v dot products first

$$s = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = \cancel{m_1^2} + \cancel{m_2^2} + 2p_1 \cdot p_2$$

- tendency forget?: diff 4-v dotted also invariant – can view as item to compare in both frames

$$E_b \quad \xrightarrow{\vec{p}_1 = E_b \hat{z}}$$

$$\xleftarrow{\vec{p}_2 = -E_b \hat{z}} E_b$$

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$$s = p_1^2 + p_2^2 + 2 p_1 \cdot p_2 = \cancel{m_1^2} + \cancel{m_2^2} + 2 p_1 \cdot p_2$$

$2[(m, 0) \cdot (E, \mathbf{p})]$
(in f.t. frame)

$$\begin{aligned} 2 \left([E_b, E_b \hat{z}] \cdot [E_b, -E_b \hat{z}] \right) &= 2(E_b^2 - \vec{p}_1 \cdot \vec{p}_2) \\ &= 2(E_b^2 + E_b^2) = 4 E_b^2 \end{aligned}$$

$$E_b \quad \vec{p}_1 = E_b \hat{z}$$

$$\vec{p}_2 = -E_b \hat{z} \quad E_b$$

Fixed Target Frame

- let's just re-iterate this one more time : in Fixed target frame:

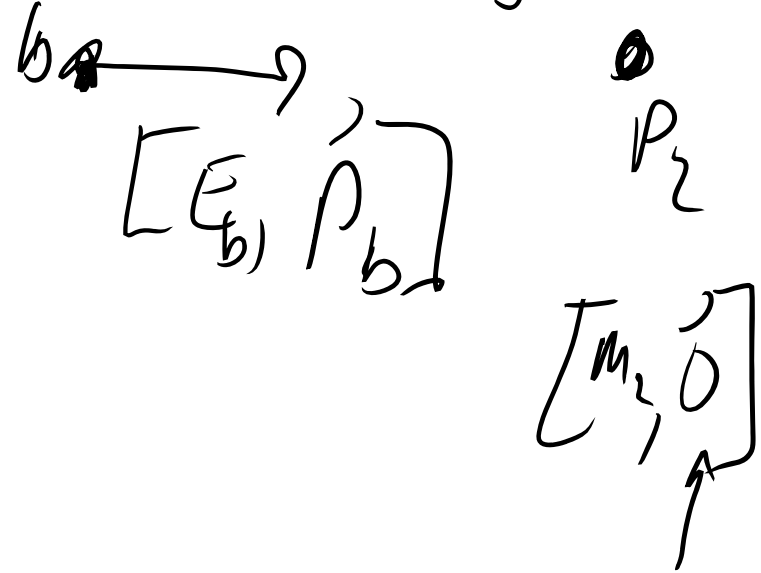
$$s = (p_1 + p_2)^2$$

$$= \cancel{p_1^2} + \cancel{p_2^2} + 2p_1 \cdot p_2$$

$m_1^2 \quad m_2^2$

$$= 2(E_b m_2 - \vec{p}_b \cdot \vec{0}) = 2E_b m_2$$

$$\sqrt{s} = \sqrt{2 E_b m} \quad \sim \quad 2 E_{\text{coll, de. b}}$$



IMPLICATION!

- Most important: lab frame vs fixed target
- Large advantage for collider:
- **MUCH HIGHER Ebeam** needed for single beam fixed target for the same COM energy

Fixed Targets vs. Colliding Beams

- In BaBar, $(9 + 3.1) = 12.1$ GeV of beam energy leads to $(2 \times 5.3) = 10.6$ GeV of CM energy that can be used to make new particles (in this case, the $\Upsilon(4S)$).
- How much beam energy would it take to produce this CM energy if the target were fixed?
- Use the total four-momentum as an invariant. The individual four-momenta will be $(m, 0)$ and $(E, \mathbf{p})$, and therefore

$$\begin{aligned}(p_{e^-} + p_{e^+})^2 &= m^2 + m^2 + 2[(m, 0) \cdot (E, \mathbf{p})] \\ &\simeq 2Em\end{aligned}$$

With $2Em = 4E_{CM}^2$ we find that $E = 10^5$ GeV!

IMPLICATION!

- $s = 2mE_b$ fixed target $\rightarrow \sqrt{s} = \sqrt{2mE_b}$
- $s = 4E_b^2$ symmetric collider $\sqrt{s} = 2 E_b$
- As expected you get twice E_b COM energy @ collider
- ...but only $\propto \sqrt{E_b}$ for fixed target
- For protons already big effect
 - ($m = 1$)
- Good example : ~Highest energy cosmic rays (UHECR)
- $\sim 10^{18} - 10^{19} \text{ eV} = 10^9 - 10^{10} \text{ GeV} \rightarrow$ protons
 - $\rightarrow \sqrt{s} \rightarrow 10^5 \text{ GeV} = 40-140 \text{ TeV}$ $\sqrt{s}$
 - $\rightarrow O(\text{LHC } 14 \text{ TeV}) \quad \text{SSC } 40 \text{ TeV} \quad \text{FCC}$
- But if they are iron nuclei (likely for the highest)
 - $\sqrt{s} \rightarrow \sqrt{s_{NN}} \rightarrow \times 1/A = 1/55$
 - $\rightarrow \sqrt{s_{NN}} \sim 800 \text{ GeV} - 2 \text{ TeV}$ LHC HI !!! (RHIC also relevant!)

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Even bigger for electron beams

- $s = 2mE_b$ fixed target $\rightarrow \sqrt{s} = \sqrt{2mE_b}$
- $s = 4E_b^2$ symmetric collider $\sqrt{s} = 2 E_b$
- Can't ignore the m for e.g. electron beams which makes the effect even bigger from $1/0.0005$ factor
- ... 10 GeV collider SLAC $\rightarrow$ 100 TeV beam needed for fixed target (5 GeV b)
- 2 TeV collider (Fermilab) $\rightarrow$ 39 GeV fixd. targ. program

- Use the total four-momentum as an invariant. The individual four-momenta will be $(m, 0)$ and $(E, \mathbf{p})$, and therefore

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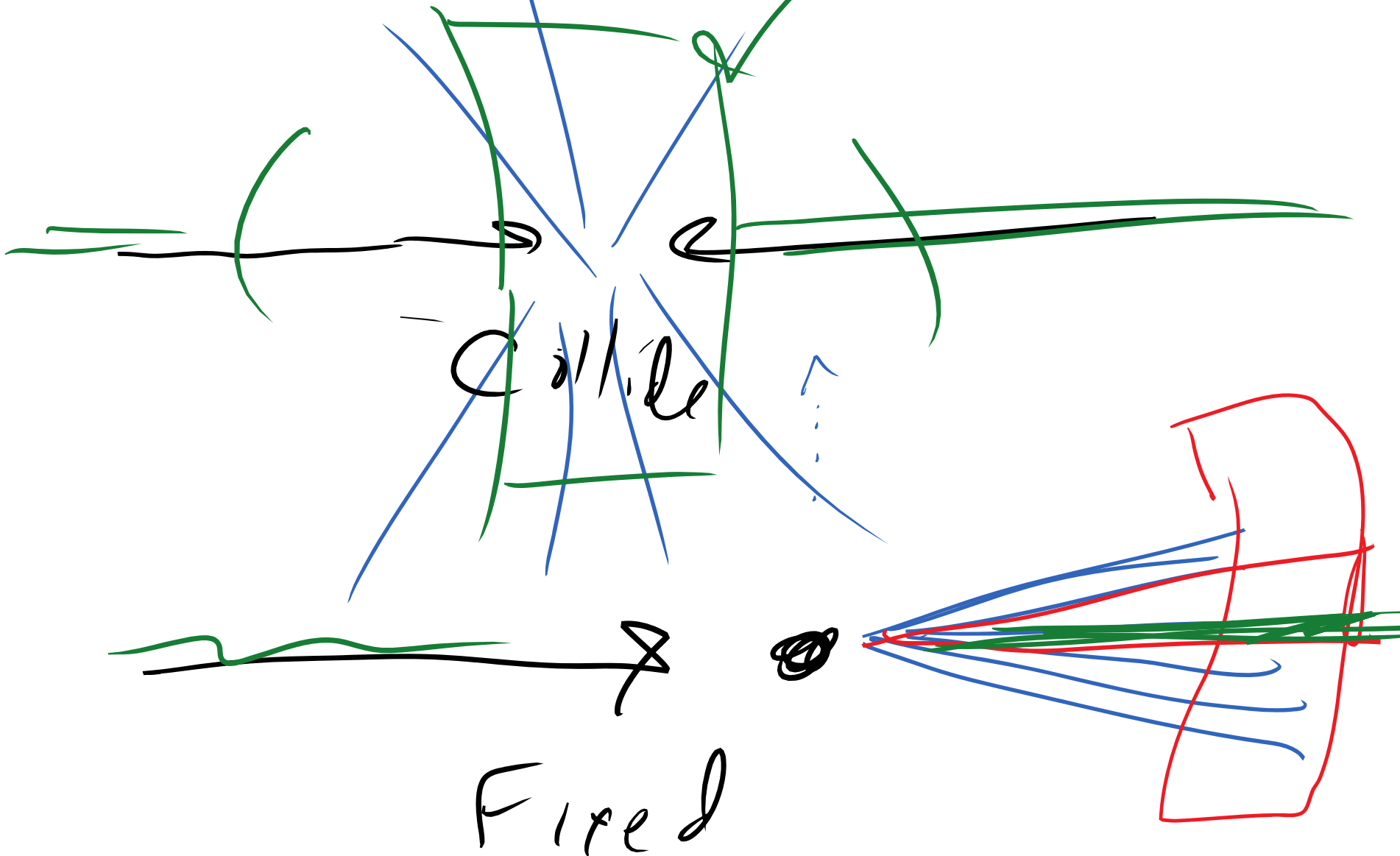
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 - $\sqrt{s} \rightarrow \sqrt{s_{NN}} \rightarrow \times 1/A = 1/55$
 - $\rightarrow \sqrt{s_{NN}} \sim 800 \text{ GeV} - 2 \text{ TeV LHC HI !!! (RHIC also relevant!)}$
- If electron CR's $\rightarrow 10^9 \text{ GeV} \rightarrow 10^{4.5} \cdot 10^{-1.5} \rightarrow 1 \text{ TeV CoM !}$

Fixed Target vs Collider



Asymmetric Beams

Evidently, even “a little” boost w.r.t. fixed already gains a lot

Fixed Targets vs. Colliding Beams

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- How much beam energy would it take to produce this CM energy if the target were fixed?

- Transform asymmetric beams

- BaBar – PEP II – SLAC

- $\text{sqrt} [([9+3, 9-3])^2] = [E^2 - p^2] \text{sqrt}(144 - 36))$

- HERA 27 GeV e x 920 GeV p

- $\text{sqrt}[27+920, 893]^2$ instead $\text{sqrt}(2p_1 \cdot p_2) = \text{sqrt}(2(27 \cdot 920))$

- EXERCISE:

- EIC : eA 10 GeV e x 100 GeV p,n, N (“ $\sqrt{s_{NN}}$ “)

- ep 18 GeV e x 250 GeV p

Exercise

- Figure out $\sqrt{s}$ for HERA Collider:
- $e+p$ electron beam 27 GeV and proton 920 GeV
- Also for EIC : do e 18 GeV, p 200 GeV

- Also for EIC : do e 18 GeV, p 200 GeV $\leftarrow$ 100 GeV at first

Remember $m^2 = E^2 - p^2 \dots = (p^\mu)^2$

similarly $p_1 \cdot p_2 = E_1 E_2 - \vec{p}_1 \cdot \vec{p}_2$

Assume $E \approx p$ + remember $\vec{p}_e \cdot \vec{p}_p$ is negative $\rightarrow \leftarrow$

Then $\sqrt{s} \approx \sqrt{2 p_1 \cdot p_2} = \sqrt{2(E_1 E_2 + p_1 p_2)} = \sqrt{4 E_1 E_2}$

For Hera $\sqrt{s} = 320$ GeV For EIC $\sim 80-100$ GeV

Accelerator Facilities Part 1

(in reverse chrono order)

- FCC (Future Circular) 100 TeV p+p ?
- EIC @ BNL - ~ 200 [p/A] x ~ 15 [e] GeV
 - (Chinese EIC?!? LHCe ??)
- LHC @ CERN (same tunnel as previous LEP) A+A (5 TeV) - p+p (13 TeV) Higgs Discovery/ QGP confirmation > 2000's (data 2010)
- RHIC @ BNL A+A - p+p QGP Discovery (2000's to current)
- CEBAF @ Jlab Multipurpose (90's - current)
- SSC (cancelled) would've been 40 TeV p+p (late 80-90's)
- HERA @ DESY 1 TeV p x 40 GeV e Multipurpose (90's)
- TEVATRON @ FNAL p+pbar 2 TeV Top Quark Discovery ('96)
- SLC @ SLAC Linear e+e- 90 GeV Multipurpose (80's-2000's)
- LEP @ CERN e+e- 209 GeV (2 W bosons) (80's/90's)
- SppS @ CERN 550 GeV p+pbar W,Z Discovery (80's)
- Main Ring @ FNAL 400 GeV p Bottom Discovery (60's-70's)
- Linac @ SLAC Linear e+ 40 GeV Charm Co-Discovery (60's)
- Bevatron/Bevalac @ LBNL - Early discoveries & first HIC ($\rightarrow$ 60's)
- AGS/Cosmotron @ BNL - Strange Early Discovery (50's)

Accelerator Facilities Part 1

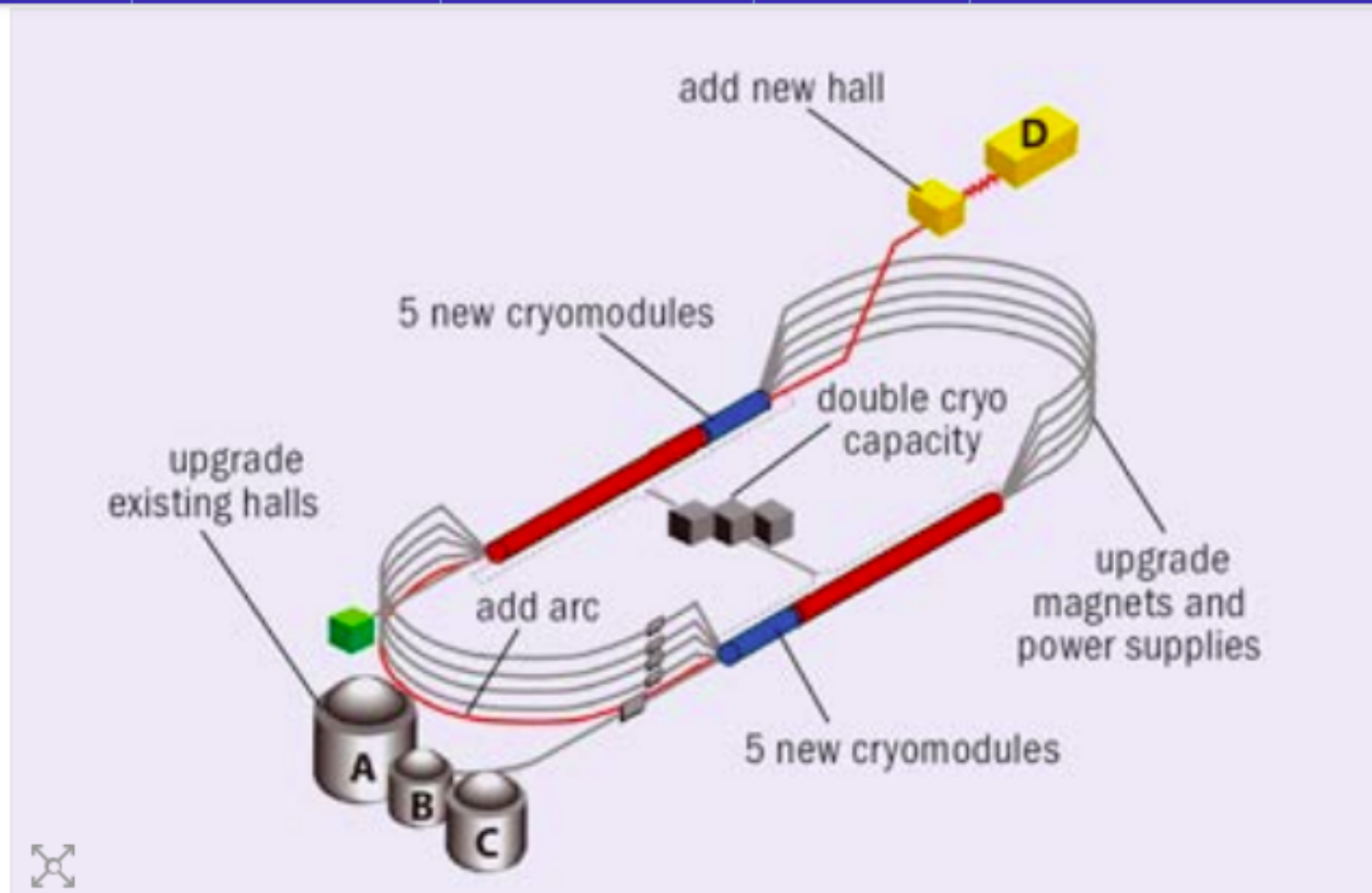
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Jlab:

not a collider, single beam accelerator

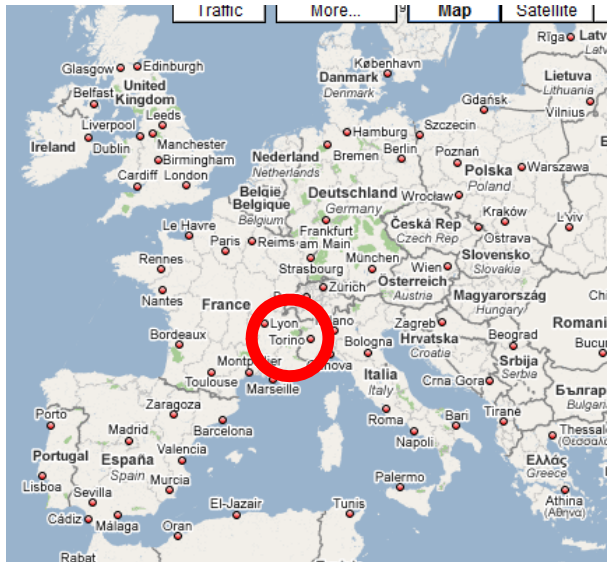
$E_{\text{beam}} = 6 \text{ GeV} \rightarrow 12 \text{ GeV}$



Schematic of the CEBAF configuration at Jefferson Lab, illustrating components of the 12 GeV

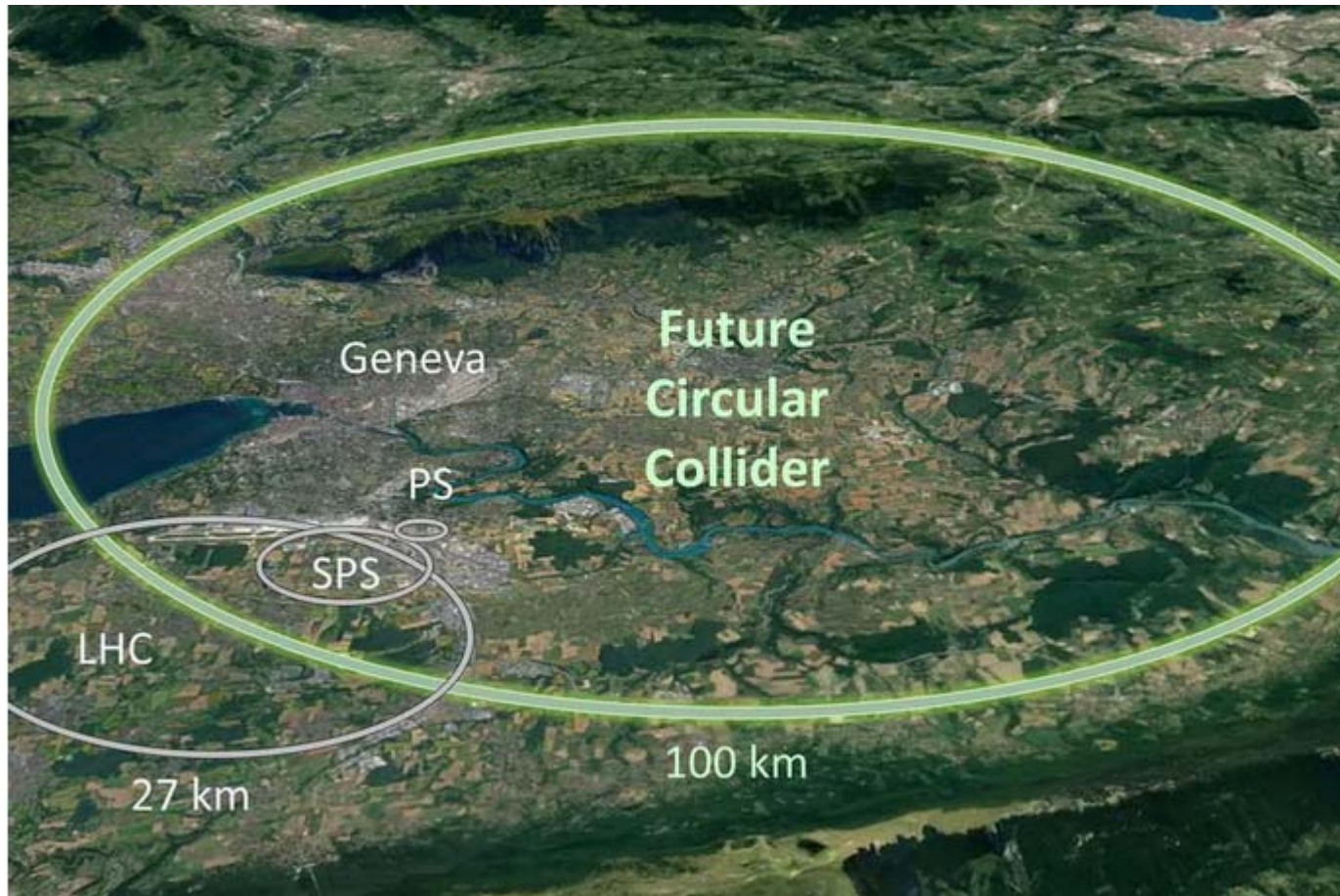
The LHC

- LHC = Large Hadron Collider – Energy Frontier
 - CERN – Switzerland/France
- LHC is the biggest particle collider ever built
- The LHC Heavy Ion program fully underway)



FCC

- Really? After SCC cut, seems unlikely in near future, but perhaps after long time...



Some Experiments/Detectors

- EIC @ BNL - ~ 200 [p/A] x ~ 15 [e] GeV (reproducibility achieved no through repeats elsewhere but at same facility)
 - **ATHENA - ECCE - CORE**
- LHC @ CERN (same tunnel as previous LEP) A+A (5 TeV) pp 13 TeV
 - **ATLAS CMS LHCb ALICE (RHI dedicated)**
- RHIC @ BNL A+A - p+p QGP Discovery (2000's to current)
 - **STAR (s)PHENIX PHOBOS BRAHMS**
- CEBAF @ Jlab Multipurpose (90's - current)
 - **Halls A- D Many experiments Hall B "CLAS" - MOELLER + ?**
- HERA @ DESY 1 TeV p x 40 GeV e Multipurpose (90's)
 - **ZEUS HERMES H1 HERA-B**
- TEVATRON @ FNAL p+pbar 2 TeV Top Quark Discovery ('96)
 - **DØ CDF MINOS BoONE (neutrino experiments) DUNE!!!**
- SLC @ SLAC Linear e+e- 90 GeV Multipurpose (80's-2000's)
 - **SLD BaBar (sPHENIX magnet provider)**
- LEP @ CERN e+e- 209 GeV (2 W bosons) (80's 90's)
 - **OPAL ALEPH DELPHI (also jets characterization)**
- SpS @ CERN 550 GeV p+pbar W,Z Discovery (80's)
 - **UA1 UA2**
- **OTHER: AMS (Alpha magnetic spectrometer) - Similar experiment located in space - spokesperson Sam Ting (J/Psi discovery)**

Back to our exercise Final State?

- $m_B \rightarrow$ defines Threshold $E \rightarrow$ In the COM:
- Sorry for confusion in this slide I started mixing the general case of threshold for 1 particle $a+a \rightarrow B$ and exercise $a+a \rightarrow B+B$.

$$p_1 \cdot p_2 = m_B^2$$

$$p_2 = p_1 = \begin{pmatrix} m_B \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \leftarrow$$

$$\sqrt{s} = \cancel{2} \sqrt{2} m_B$$

$$p_1^2 = m_B^2$$

$$\boxed{= 2 E_{\text{com}} = m_B}$$

This meant
 $E_{\text{beam-in-com}}$
 (Ea in com)

Should have
 2 here also
 for exercise
 threshold
 ($aa \rightarrow BB$)

What about final state in Lab (FT) Frame

- (for exercise $a+a \rightarrow B+B$)
- p_1/p_2 COM $\rightarrow$ for B in COM these p 's are at rest ($mB, 0, 0, 0$)
- In Fixed Target lab frame, they will be boosted by comparison
 - (NOTE: NOT same boost of γ, β of FT Beam particles !!!!!)
- Time dilation effects / collimation effects from this boost can be useful sometimes
- Leads to idea of purposely use some asymmetry in beam E 's to get little net boost?
- (not why EIC $e+p$ are asymmetric)

Dot products: $p^2 = \text{Mass}^2$ Most Important

- This relation is the most important 4-vector dot product
- $p \circ p = p^2 = m^2$
- Any momentum in ME bigger than m_{electron}
- ...and m_{pion} and m_{muon}
 - (thus can always ignore m / esp m^2)
- ...but not $m_{J/\Psi}$ (3 GeV) or sometimes not proton

Fundamental

Gauge bosons (force carriers)	str	Gluons	g	Self	1	0	0	0	0	0	0	0	Stable	
	em	Photon	γ	Self	1	0	0	0	0	0	0	0	Stable	
	w,em	W	W^+	W^-	1	80.385×10^3	0	0	0	0	0	3×10^{-25}	$e\nu_e, \mu\nu_\mu$	
	w	Z	Z^0	Self	1	91.19×10^3	0	0	0	0	0	3×10^{-25}	$e^+e^-, \mu^+\mu^-$	
Higgs boson	w, str	Higgs	H^0	Self	0	125×10^3	0	0	0	0	0	1.6×10^{-22}	$b\bar{b}, Z^0Z^0$	
Leptons	w, em ¹	Electron	e^-	e^+	$\frac{1}{2}$	0.511	0	+1	0	0	0	Stable		
		Neutrino (e)	ν_e	$\bar{\nu}_e$	$\frac{1}{2}$	$0 (<0.14 \text{ eV}/c^2)^1$	0	+1	0	0	0	Stable		
		Muon	μ^-	μ^+	$\frac{1}{2}$	105.7	0	0	+1	0	0	2.20×10^{-6}	$e^- \bar{\nu}_e \nu_\mu$	
		Neutrino (μ)	ν_μ	$\bar{\nu}_\mu$	$\frac{1}{2}$	$0 (<0.14 \text{ eV}/c^2)^1$	0	0	+1	0	0	Stable		
		Tau	τ^-	τ^+	$\frac{1}{2}$	1777	0	0	0	+1	0	2.91×10^{-13}	$\mu^- \bar{\nu}_\mu \nu_\tau, e^- \bar{\nu}_e \nu_\tau$	
		Neutrino (τ)	ν_τ	$\bar{\nu}_\tau$	$\frac{1}{2}$	$0 (<0.14 \text{ eV}/c^2)^1$	0	0	0	+1	0	Stable		
Quarks	w, em, str	(see Table 32-3)												

Hadrons (composite), selected

Mesons	str, em, w	Pion	π^+	π^-	0	139.6	0	0	0	0	0	2.60×10^{-8}	$\mu^+ \nu_\mu$
			π^0	Self	0	135.0	0	0	0	0	0	0.85×10^{-16}	2γ
		Kaon	K^+	K^-	0	493.7	0	0	0	0	+1	1.24×10^{-8}	$\mu^+ \nu_\mu, \pi^+$
			K_S^0	$\bar{K}_S^0$	0	497.6	0	0	0	0	+1	0.895×10^{-10}	$\pi^+ \pi^-, 2\pi^0$
			K_L^0	$\bar{K}_L^0$	0	497.6	0	0	0	0	+1	5.12×10^{-8}	$\pi^+ e^- \bar{\nu}_e, \pi^0 \pi^0$
		Eta	η^0	Self	0	547.9	0	0	0	0	0	5.1×10^{-19}	$2\gamma, 3\pi^0$
		Rho	ρ^0	Self	1	775	0	0	0	0	0	4.4×10^{-24}	$\pi^+ \pi^-, 2\pi^0$
			ρ^+	ρ^-	1	775	0	0	0	0	0	4.4×10^{-24}	$\pi^+ \pi^0$
		and others											
		Baryons	w	Proton	p	$\bar{p}$	$\frac{1}{2}$	938.3	+1	0	0	0	0
		Neutron	n	$\bar{n}$	$\frac{1}{2}$	939.6	+1	0	0	0	0	882	$p e^- \bar{\nu}_e$
		Lambda	Λ^0	$\bar{\Lambda}^0$	$\frac{1}{2}$	1115.7	+1	0	0	0	-1	2.63×10^{-10}	$p \pi^-, n \pi^0$
		Sigma	Σ^+	$\bar{\Sigma}^-$	$\frac{1}{2}$	1189.4	+1	0	0	0	-1	0.80×10^{-10}	$p \pi^0, n \pi^+$
			Σ^0	$\bar{\Sigma}^0$	$\frac{1}{2}$	1192.6	+1	0	0	0	-1	7.4×10^{-20}	$\Lambda^0 \gamma$
			Σ^-	$\bar{\Sigma}^+$	$\frac{1}{2}$	1197.4	+1	0	0	0	-1	1.48×10^{-10}	$n \pi^-$
		Xi	Ξ^0	$\bar{\Xi}^0$	$\frac{1}{2}$	1314.9	+1	0	0	0	-2	2.90×10^{-10}	$\Lambda^0 \pi^0$
			Ξ^-	$\bar{\Xi}^+$	$\frac{1}{2}$	1321.7	+1	0	0	0	-2	1.64×10^{-10}	$\Lambda^0 \pi^-$
		Omega and others	Ω^-	$\bar{\Omega}^+$	$\frac{1}{2}$	1672.5	+1	0	0	0	-3	0.82×10^{-10}	$\Xi^0 \pi^-, \Lambda^0 \pi^-$

$\gamma c\tau \rightarrow 10^{-10}s$
~1 m

Usually inside
detector
volumes

ID particles
by track
vertex
topology

Experimental Methods for m

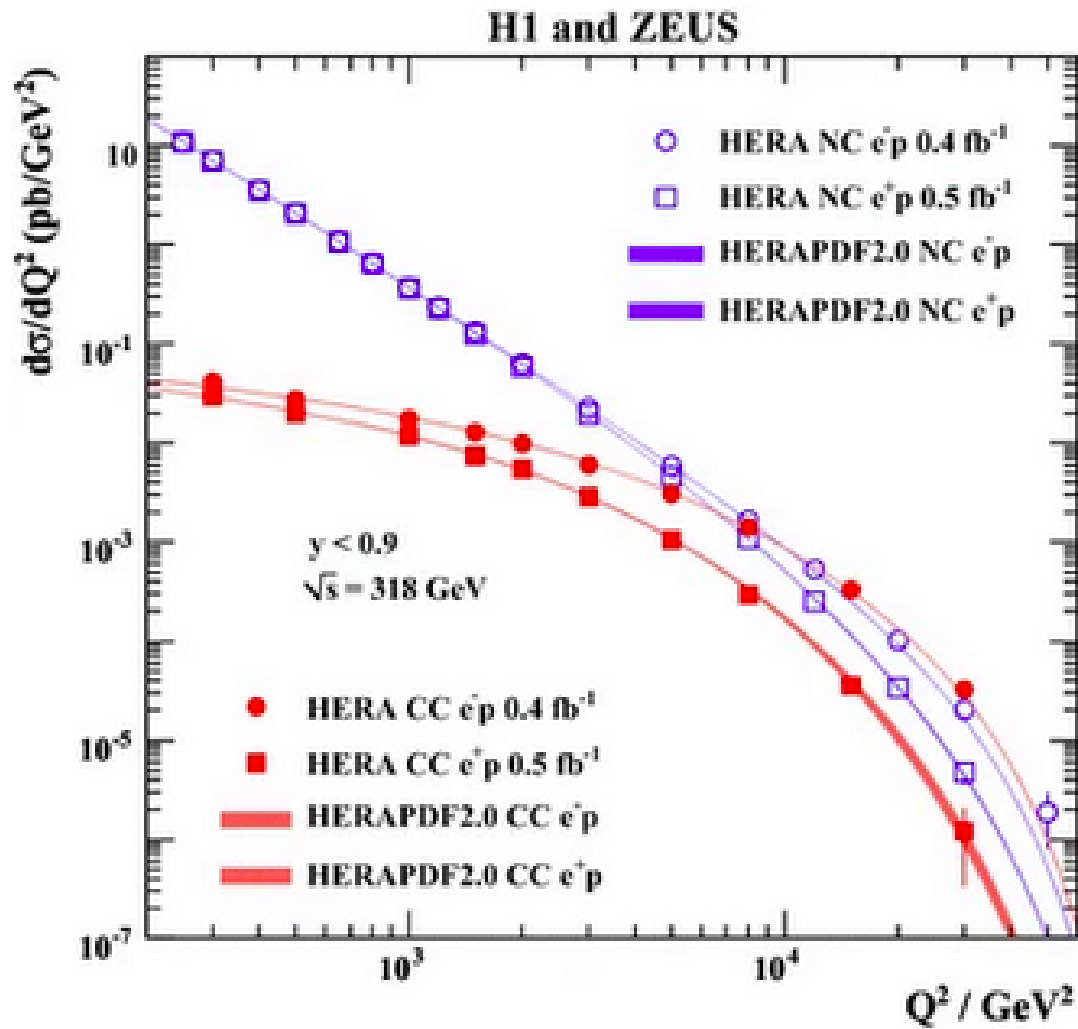
- Tracking usually gets momentum (more precise medium energy)
- Calorimeters get E
- But there are other ways, more often measure time of flight or speeds

Other Dot Products/Invariant

- How about the invariant t

t, Q^2 kine limit

- Really high $\rightarrow$ Ebeam in Fixed Target



p1-p3